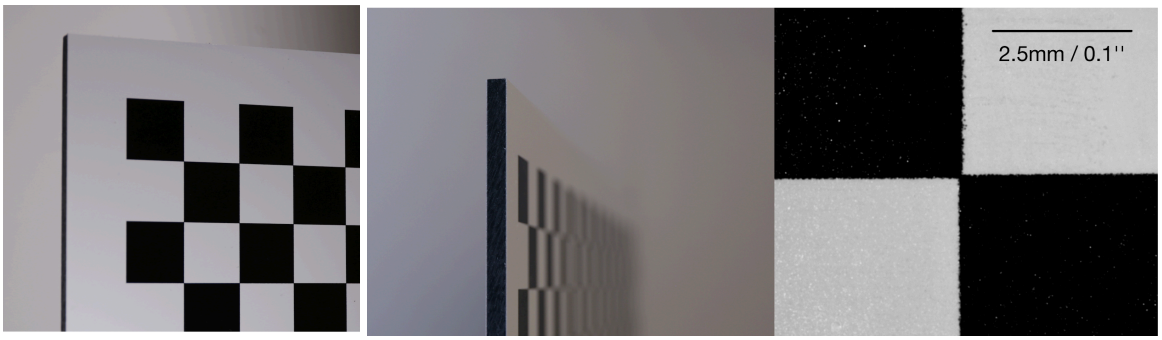
Aluminum/LDPE Calibration Targets



Optical Specifications

Patterning Process	Precision UV print
Feature Distance Tolerance	0.3mm + 0.2mm/m
Flatness Tolerance	0.5mm per 200mm
Print Resolution	1200 dpi
Gloss (@60deg)	50 (white) / 8 (black)
Gloss (@60deg) with AR coating	3 (white) / 3 (black)
Illumination (recommended)	400 - 800nm frontlit

Mechanical Specifications

Maximum Dimensions	1600 x 1200mm
Weight	6.6 kg/m²
Temperature Resistance	-50 °C to + 80 °C [-58 °F to +176 °F]
Specified temperature	20 °C (ISO 1:2016)
Linear thermal expansion	~24 µm /(°C · m)
Rigidity E·J	3840 kNcm2/m
Outdoor Lifetime	up to 3 years
Submersible	24 h (high salinity)

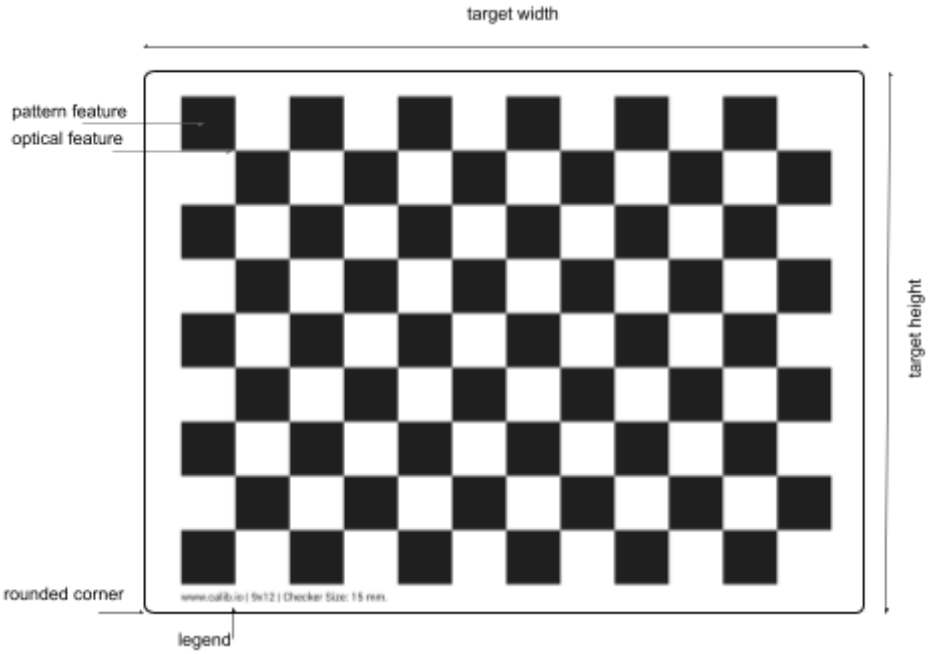
Substrate Properties

Aluminium/LDPE composite boards offer excellent flatness and stiffness, while being lightweight. Additionally, the composite material offers good thermal stability and corrosion resistance.

The patterns are transferred to the composite material using a specialized precision direct UV (Ultra-Violet) printing process resulting in a durable matte finish. This combination of substrate and patterning process is ideal for factory floor calibration, robotics, UAVs and more. Custom patterns can be created with no template cost and short turnaround time.

For applications requiring very low sheen, we can offer a proprietary antireflective (AR) coating which makes reflectance close to Lambertian while still being highly durable.

For applications requiring tighter dimensional tolerances, e.g. lab and metrology applications, we recommend lithographic print and etched targets on glass or carbon fiber substrates.



Dimensional tolerances

Aluminium/LDPE targets are manufactured on specialized precision printers that are carefully calibrated for the best possible dimensional accuracy. The manufacturing tolerance on this process is 0.3mm + 0.2mm/m, i.e. for two optical features spaced 1m apart, the maximum permissible distance error is 0.5mm.

Customization

Aluminium/LDPE targets can be customized to customer requirements. Lead times are generally between 2 days and 2 weeks.

- Custom patterns. Provided as 100% scale PDF complete vector graphics with 10mm margin to target edges. Drawing services can be offered. Ex:
    - ChAruCo targets with different dictionaries and/or marker ids, enabling multiple boards to be detected simultaneously in the same scene.
    - Circle/checkerboard targets with unique finder patterns, enabling partial observation and stereo-calibration.
    - Deltille grid patterns and other less common targets.
    - Single AruCo marker plates for VR and robot calibration purposes.
- Drilled holes and/or shapes. Provided as colored vector lines/curves in PDF drawing.
- Customized legend text and/or logos.
- Antireflective (AR) coating for very matte, durable reflective properties.

Inspection and certification

Calibration targets are carefully inspected before shipping.

The following additional certifications can be purchased:

1. Calib.io Certification. We perform several distance and flatness measurements and certify that our manufacturing and product specifications are being met.
2. Measurement Report: The absolute 3D position of each saddle point is measured and provided in a measurement report and as a digital file. For targets up to 400mm, a calibrated optical coordinate measurement machine (OMM) is used in an ISO-17025 accredited measurement laboratory. For larger targets, we use a custom measurement setup. Using actual 3D (x,y,z) positions of the saddle points can significantly improve calibration results.

Please contact us for a quote for these certification options

Stock Part Numbers

Note: for updated stock information, please visit our product pages. Current real time stock levels are shown below “Add to Cart”.

Pattern Type	Dimensions [mm]	Pattern Count	Optical Features	Weight [kg]	P/N
Checkerboard	200 x 150	18 x 25	17 x 24	0.2	CHT200150F
	200 x 150	9 x 12	8 x 11	0.2	CHT200150C
	300 x 200	18 x 29	17 x 28	0.4	CHT300200F
	300 x 200	9 x 14	8 x 13	0.4	CHT300200C
	400 x 300	18 x 25	17 x 24	0.8	CHT400300F
	400 x 300	9 x 12	8 x 11	0.8	CHT400300C
	600 x 400	18 x 29	17 x 28	1.7	CHT600400F
	600 x 400	9 x 14	8 x 13	1.7	CHT600400C
	800 x 600	18 x 25	17 x 24	3.4	CHT800600F
	800 x 600	9 x 12	8 x 11	3.4	CHT800600C

Pattern Type	Dimensions [mm]	Pattern Count	Optical Features	Weight [kg]	P/N
ChArUco	200 x 150	17 x 24	16 x 23	0.2	CCT200150F
	200 x 150	9 x 12	8 x 11	0.2	CCT200150C
	300 x 200	17 x 28	16 x 27	0.4	CCT300200F
	300 x 200	9 x 14	8 x 13	0.4	CCT300200C
	400 x 300	17 x 24	16 x 23	0.8	CCT400300F
	400 x 300	9 x 12	8 x 11	0.8	CCT400300C
	600 x 400	17 x 28	16 x 27	1.7	CCT600400F
	600 x 400	9 x 14	8 x 13	1.7	CCT600400C
	800 x 600	17 x 24	16 x 23	3.4	CCT800600F
	800 x 600	9 x 12	8 x 11	3.4	CCT800600C

Pattern Type	Dimensions [mm]	Pattern Count	Optical Features	Weight [kg]	P/N
Kalibr	400 x 300	7 x 10	14 x 20	0.8	KLB400300
	600 x 400	6 x 10	12 x 20	1.7	KLB600400
	800 x 600	7 x 10	14 x 20	3.4	KLB800600